FIN 412: THEORY OF CORPORATE FINANCE

UTILITY THEORY

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COURSE RESOURCES

- **The lecture is based on the recommended text for the course**
- **Main reading:**

- Olowe, R. A. (2017). Financial Management: Concepts, Financial System and Business Finance. 4th edition. Lagos: Brierly Jones (ORA)
- Copeland T., and Weston J. (1983). Financial Theory and Corporate Policy. Latest edition. (CTW)
- Brealey/ Myers. Principles of Corporate Finance.10th Edition (or earlier). (BM)
- Frantz/Payne/Favilukis. Study Guide. Corporate Finance. First Edition. 2011. (FPF)

COURSE RESOURCES

- **Supplementary reading**:

- Ross S., R.Westerfield, J.Jaffe. (1999). Corporate Finance. Fifth Edition. IRWIN-McGraw-Hill.
- Damodaran A. (1999). Applied Corporate Finance. Wiley&Sons.
- Daniel, K., & Marshall, D. (1997). The equity premium puzzle and the risk-free rate puzzle at long horizons'. Macroeconomic Dynamics l, 452-484.
- DeBondt, W., & Thaler, R. (1985). Does the Stock Market Overreact?, *Journal of Finance*, 40, 793-805.
- Dimson, E., & Mussavian, M. (2000). Market Efficiency, The Current State of Business Disciplines, vol.3, Spellbound Publications, 2000, pp 959-970
- Fama, E.F. (1998). Market Efficiency, Long-Term Returns, and Behavioral Finance. *Journal of Finance*, 49(3) 283-306.
- Fama, E. F., & French, K. (1992).The Cross-Section of Expected Returns, *Journal of Finance*, 47, 427- 465.
- Fama, E. F. (1970). Efficient Capital markets: A Review of Theory and Empirical Work, *Journal of Finance*, 25 (2), 383-417

CONTENT

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- Utility Functions
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Overview
(CTW– Chap. 3)

- Every individual or investor make choices everyday
- Many times, the choices are among risky alternatives, thus, uncertain
- Utility theory is the theory of choice by individuals when faced with uncertainty.
- The individual make choices among risky alternatives in such a way as to obtain maximum satisfaction

Axioms or Assumptions of Utility Theory (CWT Chap. 3)

- To develop utility theory, it is necessary to make assumptions about individual behaviour also called axioms of cardinal utility (CWT, Chap. 3).

- **Axiom 1:** Comparability

If we have an entire set say , S of uncertain alternatives, an individual can either prefer outcome x to outcome y i.e. ($x \succ y$) or y is preferred to x ($y \succ x$) or the individual is indifferent as to x and y ($x \sim y$).

Axioms or Assumptions of Utility Theory (CWT Chap. 3)

- **Axiom 2:Transitivity (also called called consistency).**
- If $x \sim y$ and $y \sim z$, then $x \sim z$

- **Axiom 3: Strong indepedence**

Suppose we set up a gamble such that an individual has a probability α of receiving outcome x and a probability $1-\alpha$ of receiving outcome z. This is denoted by $G(x,z:\alpha)$

If $x \sim y$, then $G(x,z:\alpha) \sim G(y,z:\alpha)$.

Axioms or Assumptions of Utility Theory (CWT Chap. 3)

Axiom 4: Measurability
If $x \succ y \succcurlyeq z$ or $x \succcurlyeq y \succ z$ then there exist a unique α, such that
$y \sim G(x,z:\alpha)$.

Axiom 5: Ranking.
If y and u lies between x and z , i.e.
If $x \succcurlyeq y \succcurlyeq z$ and $x \succcurlyeq y \succcurlyeq z$,
then following Axiom 4 we can establish gambles such that
$y \sim G(x,z:\alpha)$ and $u \sim G(x,z:\alpha_2)$,
it follows that
if $\alpha, > \alpha_{2,}$ then $y \succ u$, or if $\alpha_1 = \alpha_2$, then $y \sim u$.

Extension and Implication of the Assumptions (CWT Chap. 3)

- If we add another assumption that all investors always prefer more wealth to less, given the five axioms of utility theory discussed previously, then we can say that investors will always seek to maximize their expected utility of wealth.
- The maximization of expected utility of wealth is what is referred to as the theory of choice
- All investors will calculate the expected utility of wealth for all possible alternative choices and then choose the outcome that maximizes their expected utility of wealth.

Utility Function - Properties

- Given the Gamble between x and z in Axiom 3 , G(x, z:α) we can write expected wealth from the gamble as:

  $E(w) = \alpha\, G(x) + (1-\alpha) G(z)$

  For n gamble case:

$$E(w) = \sum_{i=1}^{n} G(x_i)\, \mathrm{Prob.}(x_i)$$

- Utility function is order preserving, that is, if Utility of x (U_x) is greater than utility of y (U_y), then $x \succ y$
- Expected utility can be used to rank combinations of risk alternatives

Utility Function - Properties

- In a two gamble case between x and z as shown previously
- E[U(w)] = αU(x) + (1- α)U(z)
- For n gamble case

$$E\left[U(x_i)\right] = \sum U(x_i)\,\mathrm{Prob.}(x_i)$$

- Another important property of cardinal utility functions is increasing or decreasing marginal utility.
- The measure of utility is called utiles

Utility Function

- Given the measure of utility in utiles, utility from various gambles or level of wealth can be compiled in a tabular format and plotted in form of a graph with wealth (w) on the horizontal axis and utility of wealth U(w) on the vertical axis.
- Whatever shape we obtain is our Utility functions
- Utility function can take various shapes as:
- $U(w) = a + bw$ (linear function)
- $U(w) = a + bw + cw^2$ (quadratic function)
- $U(w) = a + bw + cw^2 + dw^3$ (quadratic function)
- $U(w) = \ln w$ (logarithm function)

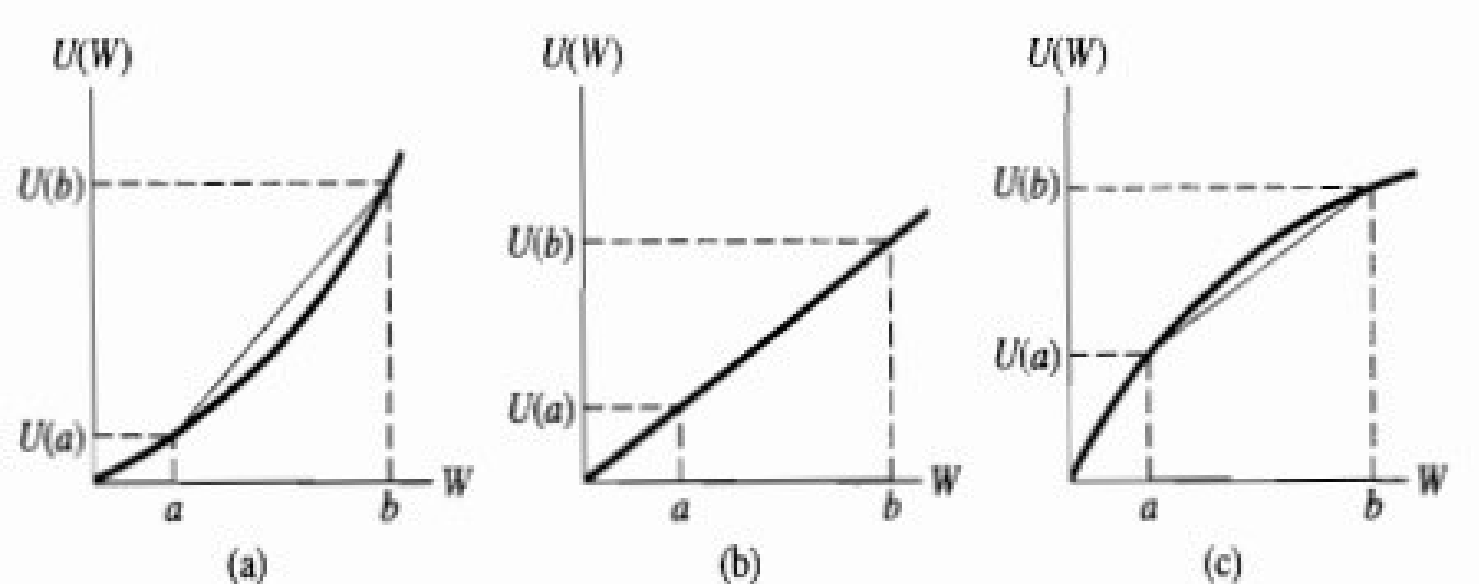
Utility Function and Risk Aversion

- The shape of an individual utility function will enable us to assess his/her attitude to risk
- Given the assumption that the individual prefer more wealth to less, that is, marginal utility of wealth is positive, there are attitudes to risk:
- Risk lover, risk neutral and risk averse

Utility Function and Risk Aversion

Figure 1: Utility Function showing three attitudes to risk: (a) risk lover, (b) risk neutral, (c) risk averse

Source: CWT, Chap. 3



Utility Function and Risk Aversion

- **From Figure 1:** The utility function of risk averse is strictly concave, while risk neutral and risk lover is convex.
- **Example 1**
  Olu, with an initial wealth of N20,000 is faced with gamble that has a 60% chance of winning N10,000 and 40% chance of winning N500. Olu has a logarithm utility function. What is his attitude to risk.
- **Solution**
- With the gamble:
- E(w) = N30,000(0.6) + N20,500(0.4) = N26,200
- Given the logarithm Utility function, U(w) = lnw
- U[E(w)]=U(N26,200) = 10.17 utiles

Utility Function and Risk Aversion

- E[U(w)] = 0.6[U(N30,000)] + 0.4[U(N20,500)] = 10.16 utiles
- Since U[E(w)] > E[U(w)], the individual is risk averse
- **Generally if:**
- U[E(w)] > E[U(w)] it indicates risk averse
- U[E(w)] = E[U(w)] it indicates risk neutral
- U[E(w)] < E[U(w)] it indicates risk lover

Utility Function and Risk Aversion
Risk Premium

- From Example 1, it is possible to calculate the maximum amount of wealth an individual would be willing to pay in order to avoid the gamble.
- This is called risk premium
- Risk premium = Expected Wealth – Certainty Equivalent wealth (CEW)
- CEW is the level of wealth that individual would accept with certainty if the gamble were removed,
- Cost of the Gamble = Current wealth - CEW

Utility Function and Risk Aversion
Risk Premium

- Thus, For Example 1, E[U(w)] =10.16 utiles
- CEW = w = $e^{10.16}$ = N25,848.30
- Risk premium = N26,200 - N25,848.30 = N351.70
- Cost of the Gamble = N20,000 – N25848.3 = -N5,848.3

Measures of Risk Aversion

- Pratt [1964] and Arrow [1971] provide a specific definition of risk aversion.
- If for instance, we have an individual with a current wealth of W presented with an actuarially neutral gamble of Z naira (by actuarially neutral we mean that $E(Z) = 0$).
- The question is to determine the risk premium, $\pi(W, Z)$, that must be added to the gamble to make the individual indifferent between it and the actuarial value of the gamble

Measures of Risk Aversion

- Mathematically,
- E[(U(w+z)] – U[E(w+z)] = $\pi(W, Z)$
- E[(U(w+z)] = U[E(w+z)] + $\pi(W, Z)$

- Using Taylor's series expansion, Pratt-Arrow of local risk premium, π, is defined as:

$$\pi = \frac{1}{2}\sigma_z^2\left(-\frac{U''(w)}{U'(w)}\right)$$

Measures of Risk Aversion

- The sign of the risk premium is determined by the items in bracket as $\frac{1}{2}\sigma^2_z$ is always positive.
- This gives rise two measures of risk aversion – Absolute Risk Aversion (ARA) and relative risk aversion (RRA)

$$\mathrm{ARA} = -\frac{U''(w)}{U'(w)}$$

ARA measures risk aversion for a given level of wealth

Measures of Risk Aversion

$$RRA = -w\frac{U''(w)}{U'(w)}$$

- If an individual has a constant relative risk aversion (RRA), it implies that he/she will have constant risk aversion to a proportional loss of wealth even though the absolute loss increases as wealth does.
- **Example 2**
- Let us now calculate the ARA and RRA for our Example 1

Measures of Risk Aversion

- U(w) = lnw

$$U'(w) = \frac{1}{w}$$

$$U''(w) = -\frac{1}{w^2}$$

$$ARA = -\frac{-1/w^2}{1/w} = \frac{1}{w} = \frac{1}{20,000}$$

- RRA = w x ARA = w x1/w = 1

SUMMARY

- Utility theory is the theory of choice by individuals when faced with uncertainty.
- To develop utility theory, it is necessary to make assumptions about individual behaviour also called axioms of cardinal utility
- Rational individuals will always maximize the expected utility of wealth

Review Question

- Try to Answer
- Suppose an individual with a logarithm utility function , U(w) = lnw and an initial wealth of N5m is exposed to a situation that results in a 50/50 chance of winning or losing N1,000,000. If the individual can buy insurance that completely removes the risk for a fee of N125,000 should he buy it or take the gamble?